Gears and wheels

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use gear ratio and wheel circumference to predict an ideal travel speed.

SUCCESS CRITERIA - I can...

1. Connect gear teeth with speed ratio.
2. Calculate distance per wheel revolution.
3. Explain the tradeoff between ideal speed and available torque.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “gear ratio” and “circumference” to explain your answer. Sentence starters are optional.

gear ratio: a comparison of gear tooth counts or rotation rates

circumference: distance around the edge of a circle

torque: the turning effect of a force about an axis

01 Warm-up: bring back an earlier idea

A **30**-turn coil has flux per turn change by **0.005 Wb** in **0.3 s**. **Find average induced-emf magnitude**.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

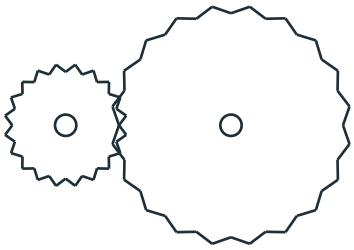
02 Worked example: explain the prediction

Reference: $rpm_{out} = rpm_{in} \times teeth_{driver}/teeth_{driven}$; $C = \pi d$
$v = C \times rpm_{wheel}/\mathbf{60}$.
Ideal gears and rolling without slip; C in meters gives v in m/s. Loading and losses change real performance.

Worked reasoning: A **10**-tooth driver turns a **30**-tooth driven gear. The driven gear makes one turn for every three driver turns. In an ideal gear pair, slower rotation trades for greater torque.

Your turn: Draw both gears, label tooth counts and opposite rotation directions.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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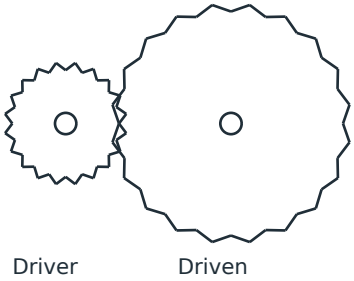
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and explain the diagram or equation

A small driver gear turns a larger driven gear. **Predict which turns more slowly** and indicate their opposite rotation directions.

Sentence starter: Both … . They differ in … because … .



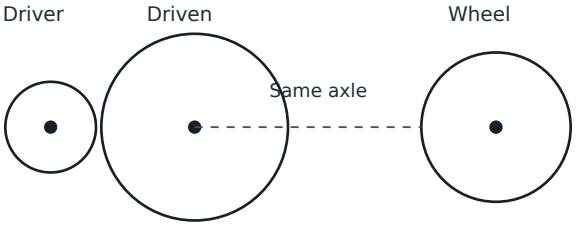
Marzano check - circle: 1 2 3

04 Apply the physics

A small classroom car motor turns at **600 rpm**. A **10**-tooth driver turns a **30**-tooth driven gear attached to a wheel of diameter **0.060 m**.
1. **Label teeth**, wheel size and rotation directions.
2. **Calculate wheel rpm** and ideal ground speed with no slip.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A team chooses the largest wheels because the formula predicts more distance per turn. **Explain why measured acceleration or** speed might still decrease.

Sentence starter: I predict … . I used … to make this prediction because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **7 Ω** resistor is in series with two parallel **10 Ω** resistors across **24 V**. **Find parallel equivalent**, total resistance and supply current.
Draw the stated circuit and **label quantities relevant** to the question.

Word help: series = connected along one unbranched current path

Sentence starter: I found … [unit]. This value means … in this situation.

Draw the complete circuit and label shared nodes.

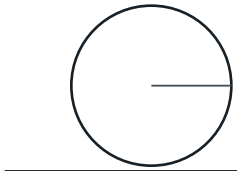
Marzano check - circle: 1 2 3

07 Exit check

A **12**-tooth driver turns a **36**-tooth gear. **Find ideal output torque as** a multiple of input torque and output speed as a fraction of input speed.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Radius / diameter: ____

Turns per second: ____

Travel per turn: ____ Ground speed: ____

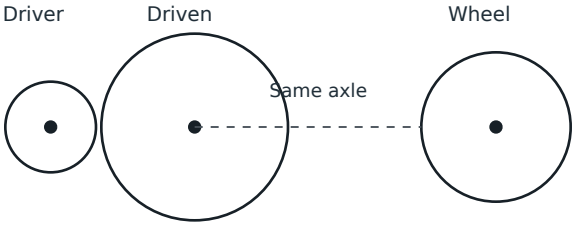
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **12**-tooth driver at **900 rpm** turns a **36**-tooth gear attached to wheels of radius **0.03 m**. **Find ideal wheel rpm** and ground speed without slip.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label teeth, rotation rates and wheel size.

Marzano check - circle: 1 2 3